Dingjin Chu (初鼎晋)

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Natural Resource Ecology and Management
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Research Interests

Forest Functions, Forest structures, Forest disturbances
Remote Sensing, Forest inventory, Process-based model, Machine learning

Academic positions

Graduate research assistant, Department of Natural Resource Ecology and Management, Oklahoma State University, Stillwater, Oklahoma, United States, June 2025 – Present

- Estimate the forest above ground biomass using LiDAR
- Assess the cooling effect of urban green space

Research assistant, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, July 2023 – May 2025

- Distinguish global forest disturbances from forest degradation, deforestation, and regrowth
- Quantify the attribution of vegetation productivity variability to climate and its extremes in semi-arid regions

Educations

Ph.D. Student, Quantitative Ecology, Oklahoma State University, Stillwater, Oklahoma, United States, May 2025 – Present (4.0 cumulative GPA)
Advisor: Lu Zhai (Asst. Prof.)

M.Sc., Forestry Ecology, Beijing Forestry University, China, 2020 - June 2023
Advisor: Kangning He (Prof.)

- Assess Forest functions across various stand structures
- Explore the relationship between timber production and forest productivity;
- Identify feasible indices derived from Sentinel-2 data to distinguish different vegetation types
- Thesis: Assessing the Forest States in the Eastern Part of Qilian Mountain Using Tree Rings and MODIS

B.Sc. (Agr.), Soil & Water Conservation and Desertification Combating, Shenyang Agricultural University, China 2016 - 2020

Community services

GSO Speaker Series Committee member, Department of Natural Resource Ecology and Management, Oklahoma State University, Stillwater, Oklahoma, United States, June 2025 – Present

- Harmonizing students and faculties to contact potential speakers and organize the activities.

Volunteer in the Walk in the Forest activity to spread the forestry knowledge to the local people at The Forest Heritage Center in Beavers Bend State Park, Oklahoma, October 11th

- Sharing the forest knowledge with the public

Member, Society of American Foresters (SAF), Oklahoma State University, Stillwater, Oklahoma, United States

Member, American Geophysical Union (AGU)

Member, Society of American Foresters **(SAF)**

Skills

Computer skills: R (Proficient in tidyverse, tidymodels, terra, data.table, and other packages), Google Earth Engine (Download and process), Python, Linux (basic command for running CLM), QGIS, ArcGIS, ArcGIS pro

Language: English (IELTS:7); Chinese (native)

Field investigation: Conducted field surveys to analyze vegetation structure, assess soil properties, and study herb diversity in ecosystems including plantations, forests, shrublands, and grasslands. Collected tree rings for further analysis.

Remote sensing: Experience with Sentinel-2, MODIS, and Landsat data

Process-based models: CLM (Running), CMIP6 (Process data), Trendy (Process data)

Publications

Chu, D., He, K., Lin, S., Zuo, Y, Chen, X. (2022). Comparison of forest and shrublands in potentially afforested areas based on remote sensing and field surveys. Acta Ecologica Sinica, 42(18), 7362-7371. (doi: 10.5846/ stxb202203210685)

Chu, D., Future climate extremes enhance interannual variability of net primary production in Central Asia, Environmental research letters. (In revision)

Chen, X., Li, Y., Zuo, Y., Lin, S., **Chu, D**., He, K. (2022). Effects of stand characteristics on herbaceous diversity. *Acta Botanica Boreali-Occidentalia Sinica*, 42(08),1396-1407. (doi:10.7606/j.issn.1000-4025.2022.08.1396)

Wu, Han, Li Zhang, Honglin He, Xiaoli Ren, Mengyu Zhang, **Dingjin Chu**, and Yuanhao Zheng. (2025) "Enhanced interannual variability of the terrestrial carbon sink in China under high emissions." Environmental Research Letters

Conference Presentations

Chu, D. Zhang, L., Wu, H., He, H., Ren, X. **Impacts of future climate extremes on interannual variability of terrestrial ecosystem productivity in Central Asia based on CMIP6**. Poster at conference Actual Issues of Environmental Research for Sustainable Development in Arid Zones, August 16-17, 2023, Issyk-Kul, Kyrgyzstan

Chu, D, **Understanding the Drivers of Local-Scale Bias in Global Forest Aboveground Biomass Maps. Poster at the AGU, December 15-29, 2025 New Orleans.**

Chu, D, **The factors influencing the accuracy of global forest biomass map to estimate local carbon storage: A case study of Tirbal Nations in the U.S. Talk at the Natural Resource Conference,** April 9-10,2025, Oklahoma City.

Scholarship & Awards & Grants

2018-2019 Second class scholarship of Shenyang Agriculture University
2020-2021 First class scholarship of Beijing Forestry University
2021-2022 First class scholarship of Beijing Forestry University
2018-2019 Merit student award of Shenyang Agriculture University
2018-2019 Model student award of Shenyang Agriculture University
2024 National Olympic Sports Cup Adult Swimming Qualification Competition (8th)
2026 Afanasiev Distinguished Graduate Fellowship Award

Personal interests

Hiking, Swimming, Cycling, Climbing
Music, Movies, Reading